

IT 520: Foundations of IT

Fall 2026 · 2026-08-31 to 2026-12-14 · 4 credits

Course Catalog

Fundamentals of computer hardware, software, and data. Topics include processors, memory, storage, resource management, and discrete mathematics.

Credits: 4

Prerequisites

CS 415, CS 410C, or CS 410P, each with a minimum grade of C-.

Course Designation

undergraduate

University Requirements

Required for the Information Technology B.S. and taken in the second-year fall. Also available as an IT cognate course.

Student Outcomes

By the end of this course, students should be able to:

- Locate a workload on the machine map, placing its steps across program, runtime, operating system, CPU and GPU, memory, storage, and network, and name the layer a stated symptom must live in.
- Measure a bottleneck and classify it as compute-, memory-, or I/O-bound, citing the measurement as evidence and re-measuring after one change.
- Predict and diagnose encoded state, naming the encoding rule that explains a wrap, signedness, rounding, text-encoding, byte-order, or format defect.
- Use discrete structures as systems machinery, stating the proposition or invariant that must hold, defining the governing set and mapping, and producing an actual count with units.
- Predict data-movement cost across the memory hierarchy and defend the prediction with locality, cache behavior, address translation, storage tier, or serialization cost.
- Choose and defend a CPU parallel execution strategy, predicting the Amdahl-bounded ceiling and naming the shared-state hazard and its remedy.
- Operate a workload under failure, isolation, and cost, specifying retry, timeout, and idempotency with the invariant that makes the retry safe.
- Diagnose a production workload and defend one measured change, carrying a machine map, bottleneck evidence, a parallelism judgment, before-and-after measurement, and one cost, reliability, or security trade-off.

Letter-grade Scale

UNH Letter Grade By Percentage

Grade	Range	Grade	Range
A	100% to 94%	A-	< 94% to 90%
B+	< 90% to 87%	B	< 87% to 84%
B-	< 84% to 80%	C+	< 80% to 77%
C	< 77% to 74%	C-	< 74% to 70%
D+	< 70% to 67%	D	< 67% to 64%
D-	< 64% to 60%	F	< 60% to 0%

Credit Hour Workload Estimate

This is a 4-credit, 15-week course. Per UNH policy, students should expect approximately 180 hours of academic work across the term (45 hours per credit), or about 12 hours per week. This is a 4-credit, 15-week course. The 180-hour total is budgeted as approximately 35 hours of scheduled contact, 2 hours of supervised assessment, and 143 hours of outside work, which is about 12 hours per week. Outside work is dominated by the assignments: you are expected to run the measurement harness, record what it reports, and write down what the numbers mean.

Topics

Machine model and measurement

- The machine map: program, runtime, operating system, CPU and GPU, memory, storage, network
- Stating a hypothesis, measuring it, and classifying the dominant stretch
- Re-measurement as the evidence that a change did something

Representation and discrete models

- Integer width, signedness, wrap, and rounding
- Text encoding, byte order, and format and delimiter assumptions
- Propositions, invariants, sets, mappings, and counting as systems machinery

Memory, storage, and data movement

- The memory hierarchy, locality, and cache behavior
- Working set, address translation, and the memory budget
- Storage tiers and serialization cost

Parallelism and heterogeneous compute

- Task, data, and pipeline parallelism, and when none of them is right
- Amdahl bounds, shared-state hazards, and their remedies
- Batch size as a decision with a count behind it
- The condition under which an accelerator loses

Production systems and integration

- Retry, timeout, and idempotency, and the invariant that makes a retry safe
- Trust boundaries, permission sets, and observable fault signals
- Cost and capacity estimation with counts
- Diagnosing a tracked production workload end to end

Sections

Modality: Hybrid

Section 01, CRN 11006. MWF 8:10 a.m.-9:00 a.m., Kingsbury S145, Durham.

Course Format

Monday and Wednesday meetings are typically in person. Friday is normally a virtual learning activity. The specific Friday activity can vary by week.

Faculty

Matt Magnusson, Senior Lecturer, Computer Science

Preferred contact: email. Email: mfm2@unh.edu. Response time: within 24 hours on weekdays.

Office: Kingsbury W243. Phone: 603-862-3794.

Office hours: Monday and Wednesday 9:00–11:00 a.m., or by appointment (Kingsbury W243 or Teams).

Evaluation

Component	Percentage
Class participation	25%
Quizzes	15%
Assignments	20%
Midterm exam	20%
Final exam	20%

How You Will Be Assessed

Class participation

Types: Assigned readings, discussion boards, classroom activities, polls, and brief reflections.

Delivery: In class or through MyCourses, as assigned.

Class participation includes assigned readings, discussion boards, classroom activities, polls, and brief reflections. Credit is based on completing a genuine attempt; simply attending is not enough.

Quizzes

Types: Short individual quizzes; may include multiple choice, short answer, and brief applied problems.

Delivery: In person during a scheduled class period, or MyCourses if specified.

Short quizzes are given through the term on recent course material. They are individual work, and they are meant to keep you current rather than to test long-range recall. Timing and coverage are announced in advance in class and in MyCourses.

Assignments

Types: Individual out-of-class assignments.

Delivery: MyCourses.

Assignments are individual work completed outside of class that applies what we cover in meetings. Instructions, due dates, and grading criteria are posted in MyCourses. Assignments are graded on the reasoning and the result you submit; completing the steps is not by itself the grade.

Midterm exam

Types: Written examination; may include multiple choice, short answer, and applied problems.

Delivery: In person during a scheduled class period.

One midterm examination during the semester. The exam assesses your individual understanding of concepts and skills covered up to that point. Format and coverage will be announced in class. University policy prohibits scheduling exams during the last week of classes.

Final exam

Types: Written examination; may include multiple choice, short answer, and applied problems.

Delivery: In person during the scheduled final-examination period.

Cumulative final examination during the university final-examination period. The exam assesses individual understanding across the course. Format and coverage will be announced in class.

Reflection / Verification Notes

Because AI tools are allowed and because the course assesses judgment, students regularly submit short verification or reflection notes. These notes should answer: What did I check? What could still be wrong? Why do I trust this result enough to submit it? Where did AI help, if it did? Verification is part of the course outcome, not optional decoration.

Late Policy for Assignments

Assignments must be completed by the posted due date unless otherwise arranged in advance. Contact me before the deadline if something is going wrong; an arrangement made ahead of time is nearly always possible, and one made afterward rarely is.

Temporary academic supports

If you are dealing with an unexpected, extenuating circumstance that will keep you out of class or affect your performance for more than a day or two, you may reach out to the Dean of Students (dean.students@unh.edu) to request a letter be sent to all your faculty.

If you are required to miss significant class time, you will be provided temporary academic supports so that you can continue to make satisfactory progress in this course. Please contact your course instructor to discuss

the specific types of supports that will be implemented during your absence. Possible options you may be provided in this class include:

- Class notes from a peer
- One-on-one meetings with instructor or TA to catch-up on missed content
- Virtual office hours
- Access to recorded lectures (asynchronous)
- Remote access for class (synchronous)
- Lecture slides
- Handouts or other materials that are distributed

Please recognize that not all students will be well enough to attend classes remotely.

Grade Disputes

If you believe a grade on an assignment, quiz, or exam is incorrect, submit a written regrade request within one week of the grade being posted to MyCourses. Coursework submitted for reconsideration may be regraded in its entirety.

Learning Resources

Textbook

No required textbook. Readings required for the course will be provided through MyCourses.

Other materials

None

Technical requirements

A laptop is required for in-class work and for the assignments. The measurement harness runs in Python; setup instructions are provided in MyCourses in the first week.

Attendance

Attendance is expected at all scheduled meetings. Class participation is graded from work produced in and around class meetings, and quizzes are given during class, so missed meetings cost credit directly. If you must miss a meeting, contact me in advance where possible.

Student AI Policy

AI tools are allowed in this course and will be part of how we learn. Students may use AI tools to help explain concepts, draft text, try prompts, debug errors, interpret documentation, brainstorm approaches, and improve written explanations. However, students are responsible for everything they submit. Allowed use includes asking AI to explain an error or concept; asking AI to draft text or prompts that you then test and revise; asking AI to help summarize documentation; asking AI to suggest checks, tests, or edge cases; and asking AI to improve clarity of writing. Not allowed: submitting AI-generated work you cannot explain; submitting unchecked AI output as fact; fabricating data, citations, sources, or results; using AI to bypass required individual work on closed assessments; hiding meaningful AI assistance when disclosure is requested. The

working standard is: AI may help you produce the work, but you must verify it, understand it, and take responsibility for it.

Instructor AI Policy

I use AI tools to help develop this course, including lectures, assignments, exercises, examples, and supporting materials. AI helps me iterate on explanations, generate examples, test assignment ideas, and improve the clarity of course materials. I grade student assignments personally. I may use AI as a tool to help improve the quality and consistency of feedback, such as identifying patterns, suggesting clearer explanations, or helping ensure feedback is thorough. Final evaluation and grades are my responsibility and my judgment. This is the same standard expected of students: AI can assist the work, but the person using it remains responsible for the result.

Academic Honesty

The University Academic Integrity Policy applies in this course. Students are expected to complete independent, original work unless the instructor specifies otherwise. The policy, including the steps faculty must take when issuing a formal penalty (reduced or failing grade on an academic assessment or examination, or reduced or failing grade in the course), is published in the Student Rights, Rules, and Responsibilities. Those steps include providing students with notice of the allegations, giving them the opportunity to respond, and reporting the violation to the Office of Community Standards if a formal grade penalty is being issued.

See the [Academic Integrity Policy](#) and [Academic Integrity in the Student Rights, Rules, and Responsibilities](#).

Plagiarism: Use or submission of intellectual property, ideas, evidence produced by another person, including computer generated text or work outsourced to third parties, in whole or in part as one's own in any academic assessment without providing proper citation or attribution. In some cases, reusing one's own previous work without acknowledging or citing the original work can constitute self-plagiarism.

Cheating: Use or attempted use of any academic exercise materials, information, study aids, electronic data, AI tools, assignment/exam surrogate, or other forms of assistance without authorization.

Class Recording

Class lectures and activities may be recorded to make this course more accessible, and recordings may be made available to students presently enrolled in the class to assist in reviewing material and preparing for assessments. Recordings are to be used for educational purposes only. You may not record class activities (including audio/photographs/screenshots of instructors and students) without express permission from me and other students whose image is captured (See NH RSA-570A). If permission is granted, student-initiated recordings may only be used for educational purposes in this class and only during the period in which you are enrolled in the class; authorized student-initiated recordings may not be posted or shared in any fashion outside of the class.

University Policies

UNH provides a variety of resources to support students' academic success, health and well-being, and engagement with the campus community.

A list of student resources can be found on the [Syllabus Supplemental Resources](#) page. These resources are not required for this course but may be helpful to you during the semester.

Students with Disabilities

According to the Americans with Disabilities Act (as amended, 2008), each student with a disability has the right to request services from UNH to accommodate their disability. If you are a student with a documented disability or believe you may have a disability that requires accommodations, please contact Student Accessibility Services (SAS) at sas.office@unh.edu.

Accommodation letters are created by SAS with the student. Please follow-up with your instructor as soon as possible to ensure timely implementation of the identified accommodations in the letter. Faculty have an obligation to respond once they receive official notice of accommodations from SAS, but are under no obligation to provide retroactive accommodations.

For more information, contact SAS: www.unh.edu/sas, (603) 862-9355, 711 (Relay NH) or sas.office@unh.edu.

Students Experiencing Emotional or Mental Health Distress

Your academic success and overall mental health are very important. If, during the semester, you find you are experiencing emotional or mental health issues, please contact Psychological and Counseling Services (PACS), which provides counseling appointments and other mental health services.

Location: The Wellness Center, 4 Pettee Brook Lane. Phone (24/7): 603.862.2090 / TTY: 7-1-1.

If urgent, students may call PACS Monday–Friday, 8 a.m.–5 p.m. and schedule an urgent same-day appointment.

Students Experiencing Housing or Food Insecurity

Food, housing, and financial supports are available on the Durham campus. See Housing, Food and Financial (Basic Needs) Support. If you have questions or need to speak with someone, contact basic.needs@unh.edu or visit the Memorial Union Building.

Classroom Behavior Expectations

Understanding your rights and responsibilities is essential to fostering a respectful and inclusive campus climate, including in the classroom. The Student Rights, Rules, and Responsibilities Handbook (“S.R.R.R.”) is the University’s official compendium for the Student Code of Conduct, cornerstone student policies and expected standards of behavior. See Academic Integrity in the S.R.R.R. and the Syllabus Supplemental Resources institutional-policies list.

Confidentiality and Mandatory Reporting

The University of New Hampshire and its faculty are committed to assuring a safe and productive educational environment for all students and for the university as a whole. To this end, the university requires faculty members to report to the university’s Title IX Coordinator (Bo Zarycky, Bo.Zarycky@unh.edu, 603-862-2930/1527 TTY) any incidents of sexual violence and harassment shared by students. If you wish to speak to a confidential support service provider who does not have this reporting responsibility because their discussions with clients are subject to legal privilege, you can contact the SHARPP Center for Interpersonal Violence Awareness, Prevention, and Advocacy at (603) 862-7233 / TTY (800) 735-2964.

For more information about what happens when you report, how the university treats your information once a report is made to the Title IX Coordinator, and your rights and reporting options at UNH (including anonymous reporting options), see the Civil Rights & Equity Office. The uSafeUS® app is your college’s free, anonymous tool that helps you and your friends navigate uncomfortable situations and access support.

Help us improve our campus and community climate. If you have observed or experienced an incident of bias, discrimination or harassment, please report the incident by contacting the Civil Rights & Equity Office at UNH.civilrights@unh.edu or (603) 862-2930 voice / (603) 862-1527 TTY / 7-1-1 Relay NH. Anonymous reports may be submitted.